THE MAXIMUM NUMBER OF ELEMENTS IN A MULTIPLICATIVE SIDON SET: UPPER BOUND

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Abstract. This is a blueprint written by ChatGPT in order to let Aristotle formalize an upper bound on the largest possible subset $A \subseteq \{1, 2, \ldots, n\}$ such that all products ab with $a, b \in A$ and $a \leq b$ are distinct. I first let ChatGPT write up Erdős' 1968 proof that shows that

$$|A| \leq \pi(n) + \frac{cn^{3/4}}{(\log n)^{3/2}}$$

for all such sets A and some absolute constant c, and then asked it to optimize the constant c as much as possible. This eventually led to showing that any

$$c > 2^{3/2} \int_0^\infty e^{-x} \sqrt{(e^x - e^{-x}) \prod_{p \leq e^{2x}} \left(1 - \frac{1}{p}\right)^{-1}} dx \approx 13.1$$

is admissible, as long as n is large enough. This paper is entirely AI-generated, not independently verified, and not optimized for readability or ease of understanding. That being said, the proof is correct, as the Lean file that Aristotle produced can be found here:

https://github.com/Woett/Lean-files/blob/main/ErdosProblem425Upper.lean

1. Definitions

Definition 1. *For $x \geq 1$, let*

$$\pi(x) = \#\{p \leq x : p \text{ is prime}\}.$$

For $m \in \mathbb{N}$, define

$$P^-(1) = +\infty, \qquad P^-(m) = \min\{p : p \text{ is prime and } p \mid m\} \quad (m > 1).$$

For $x \geq 1$ and $y \geq 1$, define

$$\Phi(x, y) = \#\{m \in \mathbb{N} : m \leq x,\ P^-(m) \geq y\}.$$

A finite set $A \subseteq \mathbb{N}$ is called multiplicative Sidon if, whenever $a, b, c, d \in A$, $a \leq b$, $c \leq d$, and

$$ab = cd,$$

then $a = c$ and $b = d$.

Definition 2 (Buchstab function)**.** *The Buchstab function $\omega : [1, \infty) \to \mathbb{R}$ is the unique continuous function satisfying*

$$\omega(u) = \frac{1}{u} \qquad (1 \leq u \leq 2)$$

and

$$(u\omega(u))' = \omega(u - 1) \qquad (u > 2).$$

Definition 3. *For $x \geq 0$, put*

$$H(x) = \prod_{p \leq e^{2x}} \left(1 - \frac{1}{p}\right)^{-1},$$

where the empty product is 1. *Define*

$$I = \int_0^\infty e^{-x} \sqrt{H(x)(e^x - e^{-x})}\, dx$$

and

$$C_* = 2^{3/2} I.$$

For $2/3 \leq \alpha < 3/4$, put

$$\beta_\alpha = \alpha - \frac{1}{2}, \qquad U_\alpha = \frac{1}{2\alpha - 1}, \qquad \Omega_\alpha = 2U_\alpha \omega(U_\alpha).$$

The auxiliary parameter α will be chosen close to $3/4$ in the proof.

2. External inputs from the literature

The proof uses the following result from the literature. It can be assumed without proof. Everything else does have to be proved.

External Literature Result 1 (Buchstab estimate)**.** *This is the form of the standard Buchstab estimate used below; it follows from Montgomery–Vaughan* [1, Theorem 7.11]. *For every real $U \geq 1$ there exist constants $K_U > 0$ and $X_U \geq 2$ such that the following holds. If $x, y \in \mathbb{R}$ satisfy $y \geq 2$, $x \geq X_U$, and*

$$1 \leq u := \frac{\log x}{\log y} \leq U,$$

then

$$\left| \Phi(x, y) - \left(\frac{\omega(u)x}{\log y} - \frac{y}{\log y} \right) \right| \leq K_U \frac{x}{(\log x)^2}.$$

Lemma 1 (Mertens product upper bound)**.** *There exists an absolute constant $K_M > 0$ such that, for every real $z \geq 3$,*

$$\prod_{p \leq z} \left(1 - \frac{1}{p}\right)^{-1} \leq K_M \log z.$$

For example, this follows from Montgomery–Vaughan [1, Theorem 2.7(e)].

A proof of a statement very close to this Mertens product upper bound has been added as Lean file.

Proof. This is a computational certificate, not a result imported from the literature. It is used only for the numerical corollary with the displayed constant 13.1. The second inequality follows from

$$2^{3/2} \cdot \frac{9263}{2000} < \frac{131}{10}.$$

□

3. Self-contained consequences and combinatorial arguments

Lemma 2 (Elementary properties of the Buchstab function)**.** *For $2 \le u \le 3$,*

$$u\omega(u) = 1 + \log(u-1).$$

In particular, if $2/3 \le \alpha < 3/4$, then

$$\Omega_\alpha = 2\left(1 + \log(U_\alpha - 1)\right)$$

and

$$\lim_{\alpha \to 3/4^-} \Omega_\alpha = 2.$$

Moreover, for every compact interval $J \subseteq [1,\infty)$ there exists $L_J > 0$ such that

$$|\omega(u) - \omega(v)| \le L_J |u - v| \qquad (u, v \in J).$$

Proof. For $2 \le u \le 3$, integrate the identity $(t\omega(t))' = \omega(t-1)$ from 2 to u. Since $2\omega(2) = 1$ and $\omega(t-1) = 1/(t-1)$ on this interval, one obtains

$$u\omega(u) - 1 = \int_2^u \frac{dt}{t-1} = \log(u-1).$$

The formula for Ω_α follows from the definition $\Omega_\alpha = 2U_\alpha\omega(U_\alpha)$ and from $2 < U_\alpha \le 3$. The limit follows from $U_\alpha \to 2$ as $\alpha \to 3/4^-$.

It remains to prove the local Lipschitz assertion. On $[1,2]$ this is clear from $\omega(u) = 1/u$. Put $g(u) = u\omega(u)$. Suppose that ω is bounded on $[1,T]$. Then $g'(u) = \omega(u-1)$ is bounded on $[2, T+1]$, so g is Lipschitz on $[2, T+1]$. Since $u \mapsto 1/u$ is Lipschitz and bounded on $[1, T+1]$, the identity $\omega(u) = g(u)/u$ shows that ω is Lipschitz on $[2, T+1]$. Induction over integer intervals gives the claim on every compact subinterval of $[1,\infty)$. ∎

Lemma 3 (Consequence of Mertens' product bound)**.** *There exists a constant $K_H > 0$ such that, for all $x \ge 0$,*

$$H(x) \le K_H(1+x).$$

Proof. If $e^{2x} < 3$, then $H(x)$ is bounded by an absolute constant. If $e^{2x} \ge 3$, Lemma 1 gives

$$H(x) = \prod_{p \le e^{2x}} \left(1 - \frac{1}{p}\right)^{-1} \le K_M \log(e^{2x}) = 2K_M x.$$

Combining the two ranges gives the claim. ∎

Lemma 4 (Convergence of the integral)**.** *The integral defining I converges.*

Proof. By Lemma 3,

$$H(x) = O(1+x).$$

Also

$$e^{-x}\sqrt{e^x - e^{-x}} = e^{-x/2}\sqrt{1 - e^{-2x}} \le e^{-x/2}.$$

Hence

$$e^{-x}\sqrt{H(x)(e^x - e^{-x})} = O\left(\sqrt{1+x}\, e^{-x/2}\right),$$

which is integrable on $[0,\infty)$. ∎

Lemma 5 (Riemann-sum convergence)**.** *As* $h \downarrow 0$,

$$(e^h - 1)\sum_{r=1}^{\infty} e^{-rh}\sqrt{H(rh)(e^{rh} - e^{-rh})} \longrightarrow I.$$

Proof. Let

$$F(x) = e^{-x}\sqrt{H(x)(e^x - e^{-x})}.$$

By the convergence lemma, F is integrable on $[0, \infty)$. On every compact interval, H is bounded, nondecreasing, and has only finitely many discontinuities. Hence F is Riemann integrable on every compact interval. Moreover,

$$e^h - 1 = h + O(h^2).$$

Thus the displayed expression is the right-endpoint improper Riemann sum for F, with h replaced by $e^h - 1 \sim h$. The tails are uniformly controlled by

$$F(x) = O\left(\sqrt{1+x}\, e^{-x/2}\right).$$

Therefore the displayed convergence follows. $\square$

Lemma 6 (Parametrized factorization)**.** *Let* $n \geq 1$*, let* $1 \leq m \leq n$*, and let* $2/3 \leq \alpha < 1$*. Then there exist positive integers* u, v *such that*

$$m = uv, \qquad v \leq u, \qquad v \leq n^\alpha,$$

and such that either u *is prime or* $u \leq n^\alpha$*.*

Proof. If $m \leq n^\alpha$, take $u = m$ and $v = 1$.

Assume $m > n^\alpha$. Let w be the least divisor of m satisfying

$$w > n^\alpha.$$

Put $z = m/w$. Then

$$z < n^{1-\alpha} \leq n^\alpha.$$

If w is prime, take $u = w$ and $v = z$.

Assume that w is composite. Write

$$w = rs, \qquad 1 < r \leq s < w.$$

By the minimality of w,

$$r \leq n^\alpha, \qquad s \leq n^\alpha.$$

Since $rs = w > n^\alpha$ and $r \leq s$, one has

$$s > n^{\alpha/2}.$$

Therefore

$$rz = \frac{m}{s} \leq \frac{n}{s} < n^{1-\alpha/2}.$$

Since $\alpha \geq 2/3$,

$$1 - \frac{\alpha}{2} \leq \alpha,$$

and hence

$$rz < n^\alpha.$$

Now

$$m = s(rz).$$

Taking

$$u = \max\{s, rz\}, \qquad v = \min\{s, rz\}$$

gives the claim. $\square$

Definition 4. *Let $n \geq 1$, let $2/3 \leq \alpha < 1$, and let $1 \leq a \leq n$. A pair $(u, v) \in \mathbb{N}^2$ is called (n, α)-admissible for a if*

$$a = uv, \qquad v \leq u, \qquad v \leq n^{\alpha},$$

and either u is prime or $u \leq n^{\alpha}$.

Lemma 7 (Minimal second factor)**.** *Let $n \geq 1$, let $2/3 \leq \alpha < 1$, let $1 \leq a \leq n$, and let (u, v) be an (n, α)-admissible pair for a with v minimal among all (n, α)-admissible pairs for a.*

1. *If p is prime, $p \mid v$, and $pu \leq n^{\alpha}$, then no such p exists.*
2. *If p is prime, $p \mid u$, $u/p < v$, and $pv \leq n^{\alpha}$, then no such p exists.*

Proof. For the first assertion, write $v = pv'$. Then

$$a = (pu)v', \qquad v' < v, \qquad v' \leq n^{\alpha}, \qquad pu \leq n^{\alpha}.$$

Also $v' \leq pu$, since $v' \leq v \leq u \leq pu$. Thus (pu, v') is an (n, α)-admissible pair for a with second factor smaller than v, a contradiction.

For the second assertion,

$$a = (u/p)(pv), \qquad u/p < v, \qquad pv \leq n^{\alpha}.$$

Also $u/p \leq pv$. Thus $(pv, u/p)$ is an (n, α)-admissible pair for a with second factor smaller than v, a contradiction. $\square$

Lemma 8 (Bipartite C_4 bound)**.** *Let G be a finite simple bipartite graph with left vertex set L, right vertex set R, and edge set E. Suppose that G has no copy of $K_{2,2}$. Put*

$$s = |L|, \qquad t = |R|, \qquad e = |E|.$$

Then

$$e^2 \leq te + ts(s-1).$$

Consequently,

$$e \leq \frac{t + \sqrt{t^2 + 4ts(s-1)}}{2} \leq t + s\sqrt{t}.$$

If $s^2 \leq t$, then

$$e \leq t + s^2.$$

Proof. For $y \in R$, let $d(y)$ be its degree. Since there is no $K_{2,2}$,

$$\sum_{y \in R} \binom{d(y)}{2} \leq \binom{s}{2}.$$

Equivalently,

$$\sum_{y \in R} d(y)^2 - e \leq s(s-1).$$

By Cauchy's inequality,

$$\sum_{y \in R} d(y)^2 \geq \frac{e^2}{t}.$$

Thus

$$\frac{e^2}{t} - e \leq s(s-1),$$

and hence

$$e^2 \leq te + ts(s-1).$$

Solving the resulting quadratic inequality gives

$$e \le \frac{t + \sqrt{t^2 + 4ts(s-1)}}{2}.$$

Since $s(s-1) \le s^2$,

$$e \le \frac{t + \sqrt{t^2 + 4ts^2}}{2} \le t + s\sqrt{t}.$$

If $s^2 \le t$, then

$$t^2 + 4ts(s-1) \le t^2 + 4ts^2 \le (t + 2s^2)^2,$$

and therefore

$$e \le t + s^2.$$

□

Lemma 9 (Rankin bound for polylogarithmically smooth reciprocal tails)**.** *Fix $A > 0$, $B > 0$, and $\eta > 0$. Uniformly for $2 \le y \le (\log n)^A$,*

$$(\log n)^B \sum_{\substack{d > n^\eta \\ p \mid d \Rightarrow p \le y}} \frac{1}{d} \longrightarrow 0 \qquad (n \to \infty).$$

If $1 \le y < 2$, the sum is empty for all sufficiently large n and the same conclusion is trivial.

Proof. Assume $2 \le y \le (\log n)^A$. Put

$$\delta = \min\left\{\frac{1}{2}, \frac{1}{2\log y}\right\}.$$

Rankin's trick gives

$$\begin{aligned} \sum_{\substack{d > n^\eta \\ p \mid d \Rightarrow p \le y}} \frac{1}{d} &\le n^{-\eta\delta} \sum_{\substack{d \ge 1 \\ p \mid d \Rightarrow p \le y}} d^{-1+\delta} \\ &= n^{-\eta\delta} \prod_{p \le y} \left(1 - p^{-1+\delta}\right)^{-1}. \end{aligned}$$

Since $\delta \le 1/2$, all factors are finite. Also

$$-\log(1-z) \le \frac{z}{1-z} \le Cz \qquad (0 \le z \le 2^{-1/2})$$

with an absolute constant C, and therefore

$$\prod_{p \le y} \left(1 - p^{-1+\delta}\right)^{-1} \le \exp\left(C \sum_{m \le y} m^{-1+\delta}\right).$$

If $y \ge e$, then $y^\delta \le e^{1/2}$ and

$$\sum_{m \le y} m^{-1+\delta} \ll \frac{y^\delta}{\delta} \ll \log y.$$

If $2 \le y < e$, the product is bounded by an absolute constant. Thus the Euler product above is at most a fixed power of y, hence at most a fixed power of $\log n$.

Finally, since $y \le (\log n)^A$, either $\delta = 1/2$, or

$$\delta = \frac{1}{2\log y} \ge \frac{1}{2A\log\log n}.$$

In both cases
$$n^{-\eta\delta}$$
dominates every fixed power of $\log n$. This proves the claim. $\square$

Lemma 10 (Sifted intervals). *Fix $2/3 \le \alpha < 3/4$, $h > 0$, $\lambda > 0$, and $\varepsilon > 0$. Put*
$$\beta = \alpha - \frac{1}{2}, \qquad L_n = \lfloor \lambda \log\log n \rfloor .$$
For all sufficiently large n, the following two estimates hold for every integer r with $1 \le r \le L_n$.

First,
$$\#\Big\{m \in \mathbb{N} : e^{-rh} n^{1/2} < m \le e^{-(r-1)h} n^{1/2},\ P^-(m) \ge e^{-rh} n^{\beta}\Big\} \le (\Omega_\alpha + \varepsilon)(e^h - 1)e^{-rh}\frac{n^{1/2}}{\log n}.$$
Second,
$$\#\Big\{m \in \mathbb{N} : e^{-rh} n^{1/2} < m \le e^{rh} n^{1/2},\ p \nmid m \text{ for every prime } p \text{ with } e^{2rh} < p < e^{(r-1)h} n^{\beta}\Big\} \le (\Omega_\alpha + \varepsilon)H(rh)(e^{rh}$$

Proof. Let
$$U_\alpha = \frac{1}{2\alpha - 1}.$$
Since
$$\beta = \alpha - \frac{1}{2} = \frac{2\alpha - 1}{2},$$
one has
$$\frac{1}{2\beta} = U_\alpha, \qquad \frac{\omega(U_\alpha)}{\beta} = 2U_\alpha \omega(U_\alpha) = \Omega_\alpha.$$
For the first estimate, put
$$X_0 = e^{-rh} n^{1/2}, \qquad X_1 = e^{-(r-1)h} n^{1/2}, \qquad Y = e^{-rh} n^{\beta}.$$
Uniformly for $1 \le r \le L_n$,
$$\frac{\log X_i}{\log Y} = U_\alpha + o(1) \qquad (i = 0, 1),$$
and
$$\log Y = \beta \log n + o(\log n).$$
The Buchstab estimate gives
$$\Phi(X_i, Y) = \frac{\omega(U_\alpha) + o(1)}{\beta \log n} X_i - \frac{Y}{\log Y} + O\left(\frac{X_i}{(\log X_i)^2}\right)$$
uniformly for $i = 0, 1$ and $1 \le r \le L_n$. Subtracting the two estimates cancels the term $Y/\log Y$, and the error term is
$$o\left((X_1 - X_0)(\log n)^{-1}\right)$$
uniformly in the stated range of r. Since
$$X_1 - X_0 = (e^h - 1)e^{-rh} n^{1/2},$$
the first estimate follows.

For the second estimate, put
$$Y_0 = e^{2rh}, \qquad Z = e^{(r-1)h} n^{\beta}, \qquad M_0 = e^{-rh} n^{1/2}, \qquad M_1 = e^{rh} n^{1/2}.$$

Let m be counted by the left-hand side. Write $m = de$, where d is the part of m all of whose prime factors are at most Y_0, and where no prime factor of e is at most Y_0. Since m has no prime factor in (Y_0, Z), one has $P^-(e) \geq Z$. Thus

$$M_0/d < e \leq M_1/d.$$

Choose $\eta > 0$ so small that $\eta < 1 - \alpha$ and such that, by continuity of ω,

$$\frac{\omega(u)}{\beta} < \Omega_\alpha + \frac{\varepsilon}{4}$$

whenever

$$|u - U_\alpha| \leq \frac{2\eta}{\beta}.$$

For all sufficiently large n, uniformly for $1 \leq r \leq L_n$, for $d \leq n^\eta$, and for $M_0/d \leq x \leq M_1/d$, one has

$$\left| \frac{\log x}{\log Z} - U_\alpha \right| \leq \frac{2\eta}{\beta}.$$

Moreover $M_0/d > Z$ in this range.

Let

$$F_Z(x) = x\,\omega\left(\frac{\log x}{\log Z}\right).$$

On the compact interval of possible values of $\log x / \log Z$, the Buchstab function is Lipschitz. Since

$$\frac{\log(M_1/M_0)}{\log Z} \ll \frac{r}{\log n} \ll \frac{\log\log n}{\log n}$$

uniformly in $1 \leq r \leq L_n$, and since

$$\frac{M_0/d}{M_1/d - M_0/d} \leq \frac{1}{e^{2h} - 1},$$

we have

$$F_Z(M_1/d) - F_Z(M_0/d) \leq (\omega(U_\alpha) + O(\eta) + o(1))\,\frac{M_1 - M_0}{d}$$

uniformly for $d \leq n^\eta$ with all prime factors at most Y_0. The choice of η and the relation

$$\log Z = \beta \log n + o(\log n)$$

therefore give, by the Buchstab estimate,

$$\begin{aligned}
&\#\{e \in \mathbb{N} : M_0/d < e \leq M_1/d,\ P^-(e) \geq Z\} \\
&\leq \left(\Omega_\alpha + \frac{\varepsilon}{2}\right) \frac{M_1 - M_0}{d \log n}
\end{aligned}$$

for all sufficiently large n, uniformly for all such r and $d \leq n^\eta$. The two terms $-Z/\log Z$ in the Buchstab formula cancel when the estimates at M_1/d and M_0/d are subtracted, and the remaining Buchstab error is

$$O_h\left(\frac{M_1/d}{(\log n)^2}\right) = o\left(\frac{M_1 - M_0}{d \log n}\right)$$

uniformly.

Summing over $d \le n^\eta$ whose prime factors are at most Y_0 gives

$$\sum_{\substack{d \le n^\eta \\ p \mid d \Rightarrow p \le Y_0}} \#\{e \in \mathbb{N} : M_0/d < e \le M_1/d,\ P^-(e) \ge Z\}$$

$$\le \left(\Omega_\alpha + \frac{\varepsilon}{2}\right)(M_1 - M_0)\frac{1}{\log n} \sum_{\substack{d \ge 1 \\ p \mid d \Rightarrow p \le Y_0}} \frac{1}{d}$$

$$= \left(\Omega_\alpha + \frac{\varepsilon}{2}\right) H(rh)(M_1 - M_0)\frac{1}{\log n}.$$

It remains to discard $d > n^\eta$. Since

$$Y_0 = e^{2rh} \le e^{2hL_n} \le (\log n)^{2h\lambda}$$

for all sufficiently large n, the Rankin-bound lemma gives, with $B = 2h\lambda + 2$,

$$\sum_{\substack{d > n^\eta \\ p \mid d \Rightarrow p \le Y_0}} \frac{1}{d} = o\left((\log n)^{-2h\lambda - 2}\right)$$

uniformly for $1 \le r \le L_n$. Hence the number of counted m for which $d > n^\eta$ is at most

$$M_1 \sum_{\substack{d > n^\eta \\ p \mid d \Rightarrow p \le Y_0}} \frac{1}{d} = o\left(\frac{n^{1/2}}{\log n}\right)$$

uniformly in r, because $M_1 \le (\log n)^{h\lambda} n^{1/2}$.

Since

$$M_1 - M_0 = (e^{rh} - e^{-rh})n^{1/2},$$

the second estimate follows, after increasing $\varepsilon/2$ to ε and using $H(rh) \ge 1$. □

Lemma 11 (No $K_{2,2}$ graph attached to a multiplicative Sidon set)**.** *Let $n \ge 1$, let $2/3 \le \alpha < 1$, and let $A \subseteq \{1, \dots, n\}$ be multiplicative Sidon. Let $B \subseteq A$. For each $a \in B$, choose one factorization*

$$a = u_a v_a.$$

Form the bipartite graph G_B whose left vertices are the integers v_a, whose right vertices are the integers u_a, and whose edges are the pairs (v_a, u_a) for $a \in B$. Then G_B has no copy of $K_{2,2}$.

Proof. Suppose that G_B contains a copy of $K_{2,2}$ with distinct left vertices v_1, v_2 and distinct right vertices u_1, u_2. Then

$$v_1u_1, \quad v_1u_2, \quad v_2u_1, \quad v_2u_2$$

are elements of A, and

$$(v_1u_1)(v_2u_2) = (v_1u_2)(v_2u_1).$$

The unordered pairs

$$\{v_1u_1, v_2u_2\} \qquad \text{and} \qquad \{v_1u_2, v_2u_1\}$$

are distinct, since equality of the unordered pairs would force either $u_1 = u_2$ or $v_1 = v_2$. This contradicts the multiplicative Sidon property. □

Lemma 12 (Small second factors)**.** *Fix $2/3 \le \alpha < 3/4$. Let n be sufficiently large. Let $A \subseteq \{1, \dots, n\}$ be multiplicative Sidon. For each nonsquare $a \in A$, choose an (n, α)-admissible pair (u_a, v_a) with v_a minimal. Put*

$$A_1 = \{a \in A : a \text{ is not a square and } v_a \le n^{1-\alpha}\}.$$

Then

$$|A_1| \le \pi(n) + O(n^\alpha).$$

Proof. The number of possible left vertices is at most $n^{1-\alpha}$. Every right vertex u_a is either prime or satisfies $u_a \le n^\alpha$. Hence the number of possible right vertices is at most

$$\pi(n) + n^\alpha.$$

Enlarge the right vertex set by isolated vertices, if necessary, so that its cardinality is

$$t = \lceil \pi(n) + n^\alpha \rceil .$$

The number of left vertices is

$$s \le n^{1-\alpha}.$$

Since $\alpha \ge 2/3$,

$$s^2 \le n^{2-2\alpha} \le n^\alpha \le t.$$

The graph has no $K_{2,2}$, so the final part of the bipartite C_4 bound gives

$$|A_1| \le t + s^2 \le \pi(n) + O(n^\alpha).$$

□

Lemma 13 (Middle second factors)**.** *Fix $2/3 \le \alpha < 3/4$. Let $h > 0$, let $\lambda > 0$ satisfy*

$$\lambda h > 3,$$

and put

$$L = \lfloor \lambda \log \log n \rfloor .$$

Let n be sufficiently large. Let $A \subseteq \{1, \dots, n\}$ be multiplicative Sidon. For each nonsquare $a \in A$, choose an (n, α)-admissible pair (u_a, v_a) with v_a minimal. Put

$$A_2 = \{a \in A : a \text{ is not a square and } n^{1-\alpha} < v_a \le e^{-Lh} n^{1/2}\}.$$

Then

$$|A_2| = o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Proof. For $k > L$, put

$$A_{2,k} = \left\{a \in A_2 : e^{-kh} n^{1/2} < v_a \le e^{-(k-1)h} n^{1/2}\right\}.$$

Only k satisfying

$$e^{-(k-1)h} n^{1/2} > n^{1-\alpha}$$

occur. Hence, for occurring k,

$$e^{kh} < e^h n^{\alpha - 1/2}.$$

The number of possible left vertices in $G_{A_{2,k}}$ is

$$O_h(e^{-kh} n^{1/2} + 1),$$

and the number of possible right vertices is

$$O_h(e^{kh} n^{1/2} + 1).$$

By the bipartite C_4 bound,

$$|A_{2,k}| \ll_h e^{kh} n^{1/2} + n^{3/4} e^{-kh/2} + 1.$$

Summing over all occurring k gives

$$|A_2| \ll_h n^{\alpha} + n^{3/4} e^{-Lh/2} + \log n.$$

Since $\alpha < 3/4$,

$$n^{\alpha} = o\left(\frac{n^{3/4}}{(\log n)^{3/2}} \right).$$

Since $\lambda h > 3$,

$$e^{-Lh/2} = O\left((\log n)^{-\lambda h/2} \right) = o\left((\log n)^{-3/2} \right).$$

Therefore

$$|A_2| = o\left(\frac{n^{3/4}}{(\log n)^{3/2}} \right).$$

□

Lemma 14 (Large second factors)**.** *Let $c > C_*$. Then there exist parameters*

$$\frac{2}{3} \le \alpha < \frac{3}{4}, \qquad h > 0, \qquad \lambda > 0, \qquad \lambda h > 3,$$

such that the following holds for all sufficiently large n. Let $A \subseteq \{1, \dots, n\}$ be multiplicative Sidon. For each nonsquare $a \in A$, choose an (n, α)-admissible pair (u_a, v_a) with v_a minimal. Put

$$L = \lfloor \lambda \log \log n \rfloor$$

and

$$A_3 = \{a \in A : a \text{ is not a square and } e^{-Lh} n^{1/2} < v_a < n^{1/2}\}.$$

Then

$$|A_3| \le c \frac{n^{3/4}}{(\log n)^{3/2}}.$$

Proof. Choose c_0 such that

$$C_* < c_0 < c.$$

By Lemma 2, $\Omega_\alpha \to 2$ as $\alpha \to 3/4^-$. Since $C_* = 2^{3/2} I$, choose α with $2/3 \le \alpha < 3/4$ so close to $3/4$ that

$$\Omega_\alpha^{3/2} I < c_0.$$

By the Riemann-sum convergence lemma, choose $h > 0$ so small that

$$\Omega_\alpha^{3/2} (e^h - 1) \sum_{r=1}^{\infty} e^{-rh} \sqrt{H(rh)(e^{rh} - e^{-rh})} < c_0.$$

Choose $\lambda > 0$ such that

$$\lambda h > 3.$$

Put

$$L = \lfloor \lambda \log \log n \rfloor .$$

For $1 \le r \le L$, define

$$A_{3,r} = \left\{ a \in A_3 : e^{-rh} n^{1/2} < v_a \le e^{-(r-1)h} n^{1/2} \right\}.$$

These sets cover A_3.

Fix r with $1 \le r \le L$. If $a \in A_{3,r}$, then

$$u_a \le \frac{n}{v_a} < e^{rh} n^{1/2}.$$

If $p \mid v_a$ and

$$p < e^{-rh} n^{\alpha - 1/2},$$

then

$$p u_a < n^{\alpha},$$

contradicting the minimal second factor lemma. Hence every possible v_a is counted by the first estimate of the sifted interval lemma with

$$\beta = \alpha - \frac{1}{2}.$$

Thus, for every fixed $\varepsilon > 0$ and all sufficiently large n,

$$s_r \le (\Omega_\alpha + \varepsilon)(e^h - 1)e^{-rh} \frac{n^{1/2}}{\log n},$$

where s_r denotes the number of possible left vertices in $G_{A_{3,r}}$.

Similarly, if $p \mid u_a$ and

$$e^{2rh} < p < e^{(r-1)h} n^{\alpha - 1/2},$$

then

$$\frac{u_a}{p} < e^{-rh} n^{1/2} < v_a$$

and

$$p v_a < n^{\alpha},$$

again contradicting the minimal second factor lemma. Hence every possible u_a is counted by the second estimate of the sifted interval lemma. Therefore

$$t_r \le (\Omega_\alpha + \varepsilon) H(rh)(e^{rh} - e^{-rh}) \frac{n^{1/2}}{\log n},$$

where t_r denotes the number of possible right vertices in $G_{A_{3,r}}$.

By the bipartite C_4 bound,

$$|A_{3,r}| \le t_r + s_r \sqrt{t_r}.$$

Thus

$$|A_{3,r}| \le t_r + (\Omega_\alpha + \varepsilon)^{3/2}(e^h - 1)e^{-rh} \sqrt{H(rh)(e^{rh} - e^{-rh})} \frac{n^{3/4}}{(\log n)^{3/2}}.$$

By Lemma 3,

$$H(rh) = O_h(1 + r).$$

Hence

$$\sum_{r=1}^{L} t_r = o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Summing over $1 \le r \le L$ gives

$$|A_3| \le (\Omega_\alpha + \varepsilon)^{3/2}(e^h - 1) \sum_{r=1}^{\infty} e^{-rh} \sqrt{H(rh)(e^{rh} - e^{-rh})} \frac{n^{3/4}}{(\log n)^{3/2}} + o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Choose $\varepsilon > 0$ sufficiently small so that the coefficient in the last display is less than c. Then the claim follows. $\square$

Theorem 1. *Let $c > C_*$. Then there exists $N \in \mathbb{N}$ such that, for every $n \geq N$, if $A \subseteq \{1, 2, \ldots, n\}$ is multiplicative Sidon, then*

$$|A| \leq \pi(n) + c\frac{n^{3/4}}{(\log n)^{3/2}}.$$

Proof. Choose c' with

$$C_* < c' < c.$$

Apply Lemma 14 with c' and fix the parameters α, h, λ supplied by that lemma. Let n be sufficiently large, and let

$$A \subseteq \{1, 2, \ldots, n\}$$

be multiplicative Sidon.

Let

$$A^{\square} = \{a \in A : a \text{ is a square}\}.$$

Then

$$\left|A^{\square}\right| \leq n^{1/2}.$$

For each $a \in A \setminus A^{\square}$, choose an (n, α)-admissible pair (u_a, v_a) with v_a minimal. Since a is not a square and $v_a \leq u_a$, one has

$$v_a < u_a.$$

Put

$$L = \lfloor \lambda \log \log n \rfloor.$$

Partition $A \setminus A^{\square}$ into

$$A_1 = \{a \in A \setminus A^{\square} : v_a \leq n^{1-\alpha}\},$$

$$A_2 = \{a \in A \setminus A^{\square} : n^{1-\alpha} < v_a \leq e^{-Lh} n^{1/2}\},$$

and

$$A_3 = \{a \in A \setminus A^{\square} : e^{-Lh} n^{1/2} < v_a < n^{1/2}\}.$$

These sets cover $A \setminus A^{\square}$, because $v_a \leq u_a$ and $u_a v_a \leq n$ imply

$$v_a \leq n^{1/2},$$

and equality would force $u_a = v_a = n^{1/2}$, contradicting that a is not a square.

By the small second factors lemma,

$$|A_1| \leq \pi(n) + O(n^{\alpha}).$$

By the middle second factors lemma,

$$|A_2| = o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

By Lemma 14, applied with c' and with the parameters fixed above,

$$|A_3| \leq c' \frac{n^{3/4}}{(\log n)^{3/2}}$$

for all sufficiently large n.

Combining the estimates,

$$\begin{aligned}|A| &\le \left|A^{\square}\right| + |A_1| + |A_2| + |A_3| \\ &\le n^{1/2} + \pi(n) + O(n^{\alpha}) + o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right) + c'\frac{n^{3/4}}{(\log n)^{3/2}},\end{aligned}$$

where $C_* < c' < c$. Since $\alpha < 3/4$,

$$n^{1/2} + O(n^{\alpha}) = o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Thus, for all sufficiently large n,

$$|A| \le \pi(n) + c\frac{n^{3/4}}{(\log n)^{3/2}}.$$

□

Corollary 1. *There exists $N \in \mathbb{N}$ such that, for every $n \ge N$, if $A \subseteq \{1, 2, \ldots, n\}$ is multiplicative Sidon, then*

$$|A| \le \pi(n) + 13.1\,\frac{n^{3/4}}{(\log n)^{3/2}}.$$

Proof. A finite interval-arithmetic computation with directed rounding gives

$$I < \frac{9263}{2000}.$$

Consequently,

$$C_* < \frac{131}{10} = 13.1.$$

Apply the theorem with $c = 13.1$. □

References

[1] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I. Classical Theory*, Cambridge Studies in Advanced Mathematics, vol. 97, Cambridge University Press, 2007.